Diego Atencia

AI Engineering — LLMs · RAG · AWS · Banking Data

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Summary

M.Sc. in AI, B.Sc. in Mathematics (UAM). Experience in credit-risk (LGD/IRB) modeling and regulatory data pipelines (FINREP, COREP, CIRBE) in banking. Built and deployed regllm.xyz, a GenAI application for Spanish banking regulation (LoRA fine-tuning, RAG, AWS). Open-source contributions to LLM and RL infrastructure (Liger-Kernel, TorchRL).

Experience

Senior Data Scientist *Jun 2025 – Present*
[1] — Madrid

- Data engineering and analysis on the LGD database of a top-4 Spanish bank, where small errors have significant impact on the bank's reserved capital and earnings.
- Contributed to landing 8 contract renewals with this client; integrated 2 junior engineers now working full-time on the account.
- Developing a data-quality-control (DQC) automation tool, currently at proof-of-concept stage.

[2] *Nov 2023 – Jun 2025*
[1] — Madrid

- Built and validated LGD (Loss Given Default) credit-risk models for IRB portfolios, ensuring regulatory compliance.
- Developed Power Automate/Copilot Studio workflows integrating LGD model outputs into structured Word reporting pipelines.

SQL & PL/SQL Developer *Oct 2022 – Nov 2023*
Álamo Consulting — Madrid

- Owned regulatory reporting pipelines (FINREP, COREP, SIRBE/CIRBE) for an external banking entity using Oracle and T-SQL.
- Built scalable transformations within a large regulatory codebase, ensuring accurate and timely data delivery.

Selected Projects

regllm.xyz — GenAI app for Spanish banking regulation Q&A *Production, 2026*

- Fine-tuned Qwen2.5-7B with LoRA; RAG pipeline; full-stack deployment on AWS (ECS Fargate, RDS/pgvector, ALB).
- Built end-to-end; serving real user queries in early access (~30 to date).

Hand Prosthesis Control with Deep RL — blog *M.Sc. thesis*

- Multimodal RL agent (PPO/DDPG) integrating stereoscopic vision and sensor data for real-time prosthetic hand control in MuJoCo simulation.

Open-Source Contributions

Liger-Kernel — github.com/linkedin/Liger-Kernel PR #1128 (under review)
Production CUDA kernel library for LLM training, used at scale across industry

- Debugging a distributed-training bug in `LigerTiledSwiGLUMLP`: FSDP's post-backward reshard hooks were triggered prematurely inside a tiling loop, corrupting parameter shards across GPUs.
- Adapting axolotl's gradient-accumulator approach as the fix.

TorchRL — github.com/pytorch/rl Discussion #3538
PyTorch RL library (Meta)

- Proposing and implementing MC-PILCO: a model-based RL algorithm using Gaussian Process world models and Monte Carlo rollouts.

LEANN Issue #41

- Proposed a hybrid BM25+RRF retrieval pipeline over the neural-only architecture to improve recall on sparse queries.

Education

M.Sc. in Artificial Intelligence (8.67/10) *Sep 2024 – Oct 2025*
International University of Valencia
Thesis: *Optimizing low-cost hand prosthesis control using Deep Reinforcement Learning.*

B.Sc. in Mathematics *Sep 2018 – Jun 2023*
Autonomous University of Madrid (UAM)
Final thesis: Weak solutions to Navier–Stokes equations (9/10). Founder of university AI student association (70+ members).

Technical Skills

ML / GenAI	PyTorch, TensorFlow, Scikit-learn, HuggingFace, LoRA/QLoRA, RAG pipelines
Languages	Python (expert), SQL (expert), C/C++ (strong)
Systems & Cloud	AWS, Docker, Linux, Git/GitHub, CI/CD principles, distributed training (FSDP/DDP), CUDA kernels
Automation	Power Automate, Copilot Studio
Databases	Oracle, PostgreSQL, ChromaDB (vector), SAS

Languages

Spanish (native)   English (fluent, professional working proficiency)   Mandarin Chinese (learning)